

Smart Attendance System - Project Report

Date: 2026-05-29

Project Name: FaceTrack (Smart Face Attendance System)

Repository: <https://github.com/VanshPasricha/Xebia-Internship.git>

1. Executive Summary

FaceTrack is a smart, automated attendance system built using Python, PyQt6, and OpenCV. It replaces manual roll calls with an efficient facial recognition pipeline, significantly reducing administrative overhead and preventing proxy attendance.

2. Architecture & Database Design

The system utilizes a lightweight SQLite database to manage entities:

- Students: Stores student details and facial embeddings.
- Faculty: Manages professor credentials and roles.
- Timetable: Maps faculties to subjects, classrooms, and timeslots.
- Attendance: Records presence tied to specific timetable sessions.

3. Key Features Implemented

- Real-Time Face Recognition: Uses OpenCV to capture faces via webcam, compute embeddings, and match them against the student database.
- Role-Based Access Control: Distinct interfaces for Admin, Faculty, and Students.
- Faculty Dashboard & Analytics: Faculty can initiate attendance for ongoing classes and instantly export attendance records to CSV files.
- Student Portal: Students can securely log in to view their scheduled classes and track their attendance percentages in real-time.
- Duplicate Prevention: The system ensures a student is only marked present once per session.

4. Implementation Process

The project was developed progressively:

1. Configured the SQLite database and GUI framework.
2. Built the student registration module to capture and store face data securely.
3. Implemented the core facial recognition engine.
4. Integrated the attendance marking logic with the timetable system.
5. Added analytics dashboards and the CSV export utility for reporting.

5. Conclusion

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The Smart Attendance System successfully demonstrates the application of computer vision in educational administration. The system is scalable, user-friendly, and securely version-controlled on GitHub.